MACHINE LEARNING TYPES - SUPERVISED VS UNSUPERVISED VS REINFORCED LEARNING

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SUPERVISED LEARNING

- SUPERVISED LEARNING, YOU TRAIN THE MACHINE USING DATA WHICH IS WELL "LABELED.” → IT MEANS SOME DATA IS ALREADY TAGGED WITH THE CORRECT ANSWER.
- LEARNS FROM LABELED TRAINING DATA, HELPS YOU TO PREDICT OUTCOMES FOR UNFORESEEN DATA.

CONTOH DATASET SUPERVISED LEARNING

No	Jumlah Tanggungan Keluarga	Luas Ruma	Pendapatan/	Daya	Perlengkapan	Penggunaan
1	banyak	besar				
2	sedang	kecil				
3	sedang	stand				
4	sedikit	stand				
5	sedang	besar				
6	sedikit	besar				
7	sedang	besar				
20	sedikit	besar				

A	B	C	D	E	F	G	H	I	J	K	L	M
Loan_ID	Gender	Married	Depender	Education	Self_Empl	Applicantl	Coapplica	LoanAmou	Loan_Amo	Credit_His	Property_	Loan_Status
LP001003	Male	Yes	1	Graduate	No	4583	1508	128	360	1	Rural	N
LP001005	Male	Yes	0	Graduate	Yes	3000	0	66	360	1	Urban	Y
LP001006	Male	Yes	0	Not Gradu	No	2583	2358	120	360	1	Urban	Y
LP001008	Male	No	0	Graduate	No	6000	0	141	360	1	Urban	Y
LP001013	Male	Yes	0	Not Gradu	No	2333	1516	95	360	1	Urban	Y
LP001024	Male	Yes	2	Graduate	No	3200	700	70	360	1	Urban	Y
LP001027	Male	Yes	2	Graduate		2500	1840	109	360	1	Urban	Y
LP001029	Male	No	0	Graduate	No	1853	2840	114	360	1	Rural	N
LP001030	Male	Yes	2	Graduate	No	1299	1086	17	120	1	Urban	Y
LP001032	Male	No	0	Graduate	No	4950	0	125	360	1	Urban	Y
LP001034	Male	No	1	Not Gradu	No	3596	0	100	240		Urban	Y
LP001036	Female	No	0	Graduate	No	3510	0	76	360	0	Urban	N
LP001038	Male	Yes	0	Not Gradu	No	4887	0	133	360	1	Rural	N
LP001041	Male	Yes	0	Graduate		2600	3500	115		1	Urban	Y
LP001043	Male	Yes	0	Not Gradu	No	7660	0	104	360	0	Urban	N
LP001047	Male	Yes	0	Not Gradu	No	2600	1911	116	360	0	Semiurbai	N
LP001050		Yes	2	Not Gradu	No	3365	1917	112	360	0	Rural	N
LP001068	Male	Yes	0	Graduate	No	2799	2253	122	360	1	Semiurbai	Y
LP001073	Male	Yes	2	Not Gradu	No	4226	1040	110	360	1	Urban	Y
LP001086	Male	No	0	Not Gradu	No	1442	0	35	360	1	Urban	N
LP001087	Female	No	2	Graduate		3750	2083	120	360	1	Semiurbai	Y
LP001095	Male	No	0	Graduate	No	3167	0	74	360	1	Urban	N
LP001097	Male	No	1	Graduate	Yes	4692	0	106	360	1	Rural	N
LP001098	Male	Yes	0	Graduate	No	3500	1667	114	360	1	Semiurbai	Y
LP001109	Male	Yes	0	Graduate	No	1828	1330	100		0	Urban	N

	H	I
	Age	Outcome
527	50	1
351	31	0
572	32	1
167	21	0
288	33	1

A	B	C	D	E	F	AC	AD	AE	AF
d	diagnosis	radius_m	texture_m	perimete	area_mea	concavity_	concave p	symmetry	fractal_di
842302	M	17.99	10.38	122.8	1001	0.7119	0.2654	0.4601	0.1189
842517	M	20.57	17.77	132.9	1326	0.2416	0.186	0.275	0.08902
84300903	M	19.69	21.25	130	1203	0.4504	0.243	0.3613	0.08758
84348301	M	11.42	20.38	77.58	386.1	0.6869	0.2575	0.6638	0.173
84358402	M	20.29	14.34	135.1	1297	0.4	0.1625	0.2364	0.07678
843786	M	12.45	15.7	82.57	477.1	0.5355	0.1741	0.3985	0.1244
844359	M	18.25	19.98	119.6	1040	0.3784	0.1932	0.3063	0.08368
84458202	M	13.71	20.83	90.2	577.9	0.2678	0.1556	0.3196	0.1151
844981	M	13	21.82	87.5	519.8	0.539	0.206	0.4378	0.1072
84501001	M	12.46	24.04	83.97	475.9	1.105	0.221	0.4366	0.2075
845636	M	16.02	23.24	102.7	797.8	0.1459	0.09975	0.2948	0.08452
84610002	M	15.78	17.89	103.6	781	0.3965	0.181	0.3792	0.1048
846226	M	19.17	24.8	132.4	1123	0.3639	0.1767	0.3176	0.1023
846381	M	15.85	23.95	103.7	782.7	0.2322	0.1119	0.2809	0.06287
84667401	M	13.73	22.61	93.6	578.3	0.6943	0.2208	0.3596	0.1431
84799002	M	14.54	27.54	96.73	658.8	0.7026	0.1712	0.4218	0.1341
848406	M	14.68	20.13	94.74	684.5	0.2914	0.1609	0.3029	0.08216
84862001	M	16.13	20.68	108.1	798.8	0.4784	0.2073	0.3706	0.1142
849014	M	19.81	22.15	130	1260	0.5372	0.2388	0.2768	0.07615
8510426	B	13.54	14.36	87.46	566.3	0.239	0.1288	0.2977	0.07259
8510653	B	13.08	15.71	85.63	520	0.189	0.07283	0.3184	0.08183
8510824	B	9.504	12.44	60.34	273.9	0.08867	0.06227	0.245	0.07773

CONTOH DATASET SUPERVISED LEARNING

Tipe	Kelompok	Air	Kalori	Protein	Lemak	Karbohidr	Serat	Abu	Kalsium	Fosfor	Besi	Natrium	Kalium	Tembaga	Seng	VitaminA	BetaKarot	KarotenTc	VitaminB1	VitaminB2	Niasin	VitaminC	BDD	Sumber	Status
Olahan	Ikan dsb	11.6	513	23.7	37	21.3	0	6.4	25	25	0.5	61	0.015	0	0.2	0	0	0	0	0	0	0	100	KZGMI-OK	Cukup
Olahan	Ikan dsb	6.4	435	27.2	20.2	36.1	0	10.1	0	0	0	0	0	0	0	0	0	0	0	0.8	24.3	0	100	KZGMI-OK	Cukup
Olaha															0	0	0	0	0	0	0	0	100	PPKP-KZG	Cukup
Menta															0.2	0	83	90	0.02	0.03	1.9	2	100	KZGPI-199	Tidak
Menta															0.4	0	0	35	0.02	0.09	0.6	46	90	KZGPI-199	Cukup
Menta															0.4	0	189	180	0.05	0.08	1	13	61	DABM-KZ(	Cukup
Olaha															0	0	0	0	0	0	0	0	100	KZGMI-OK	Tidak
Olaha															0	0	0	0	0.05	0	0	0	100	KZGPI-199	Tidak
Olaha															0	0	0	0	0.07	0	0	0	100	KZGPI-199	Tidak
Olaha															2.8	0	0	0	0.2	0.1	0.5	0	100	DABM-KZ(	Tidak
																	0	0	0.06	0.02	0.1	0	100	KZGPI-199	Tidak
																	37	0	0.13	0.17	7	0	100	DABM-KZ(	Cukup
																	30	0	0.08	0.14	5.7	0	100	DABM-KZ(	Cukup
																	83	0	0.14	0.15	5.9	0	100	DABM-KZ(	Cukup
																	0	6.3	3	0	0	15	100	KZGPI-199	Tidak
																	1.717	0	0.14	0.09	0.5	10	80	DABM-KZ(	Tidak
																	0	1	0.01	0	0	3	100	KZGPI-199	Cukup
																	0	0	0.1	0.24	7.1	0	60	DABM-KZ(	Tidak
																	3.263	0	0.2	0	0	0	100	KZGMI-OK	Tidak
																	0	0	0	0	0	0	100	KZGMI-OK	Tidak
																	26	2.24	0.05	0.03	0.1	5	88	KZGPI-199	Cukup
																	30	90	0.04	0.03	0.1	5	88	DABM-KZ(	Cukup
																	0	0	0	0	0	0	100	KZGMI-OK	Cukup
																	2	60	0.03	0	0.1	60	96	DABM-KZ(	Tidak

StockCode	Description	Quantity	UnitPrice	Quality
85123A	WHITE HANGING HEART T-LIGHT HOLDER		2,55	2
71053	WHITE METAL LANTERN	6	3,39	2
84406B	CREAM CUPID HEARTS COAT HANGER	8	2,75	4
22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	3,75	5
22310	IVORY KNITTED MUG COSY			5
84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	4,25	3

Day	Outlook	Temp.	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes

UNSUPERVISED LEARNING

- UNSUPERVISED LEARNING IS A MACHINE LEARNING TECHNIQUE, WHERE YOU DO NOT NEED TO SUPERVISE THE MODEL
- IT MAINLY DEALS WITH THE UNLABELLED DATA
- ALLOW YOU TO PERFORM MORE COMPLEX PROCESSING TASKS COMPARED TO SUPERVISED LEARNING
- UNSUPERVISED LEARNING CAN BE MORE UNPREDICTABLE COMPARED WITH OTHER NATURAL LEARNING DEEP LEARNING AND REINFORCEMENT LEARNING METHODS

CONTOH DATASET UNSUPERVISED LEARNING

	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	12669	9656	7561	214	2674	1338
1	7057	9810	9568	1762	3293	1776
2	6353	8808	7684	2405	3516	7844

A	B	C	D	E	F	G	H	I	J
country	child_mort	exports	health	imports	income	inflation	life_expe	total_fer	gdpp
Afghanist	90.2	10	7.58	44.9	1610	9.44	56.2	5.82	5
Albania	16.6	28	6.55	48.6	9930	4.49	76.3	1.65	40
Algeria	27.3	38.4	4.17	31.4	12900	16.1	76.5	2.89	44
Angola	119	62.3	2.85	42.9	5900	22.4	60.1	6.16	35
Antigua ar	10.3	45.5	6.03	58.9	19100	1.44	76.8	2.13	122
Argentina	14.5	18.9	8.1	16	18700	20.9	75.8	2.37	103
Armenia	18.1	20.8	4.4	45.3	6700	7.77	73.3	1.69	32
Australia	4.8	19.8	8.73	20.9	41400	1.16	82	1.93	519
ria	4.3	51.3	11	47.8	43200	0.873	80.5	1.44	469
baijar	39.2	54.3	5.88	20.7	16000	13.8	69.1	1.92	58
mas	13.8	35	7.89	43.7	22900	-0.393	73.8	1.86	280
ain	8.6	69.5	4.97	50.9	41100	7.44	76	2.16	207
lades	49.4	16	3.52	21.8	2440	7.14	70.4	2.33	7
ados	14.2	39.5	7.97	48.7	15300	0.321	76.7	1.78	160
rus	5.5	51.4	5.61	64.5	16200	15.1	70.4	1.49	60
um	4.5	76.4	10.7	74.7	41100	1.88	80	1.86	444

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
5.1	3.5	1.4	0.2
4.9	3	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2

CONTOH DATASET UNSUPERVISED LEARNING

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	StockCode	PRODUCT	RODUCT_CATEGO	Quantity	UnitPrice	nitPriceRupia	oldCUSTID	CustomerID	CUSTNAME	ADDRESS	KOTA	PROVINSI	NEGARA	CHANNELID_SPLR	CHANNELNAME_SPLR	SUBDISTID	SUBDIST_NAME
2	85123A	123 BENDERA COKLAT 300G	SUSU	6	2.55	36465	3012815	17850	DUA PUTRI SLAMET RIYADI	JL. SLAMET RIYADI	SAMARINDA	IANTAN TIMUR	INDONESIA	32	Toko Kelontong	130113	CV. EKA PUTRA
3	71053	2.3.4 FILTER	ROKOK	6	3.39	48477	3012909	17850	SANURI	JL. M. SAID	SAMARINDA	IANTAN TIMUR	INDONESIA	32	Toko Kelontong	130113	CV. EKA PUTRA
4	84406B	234 KERETEK	ROKOK	8	2.75	39325	1921270	17850	EMI MBAK	PS. TALUN NO.63	MAGELANG	AWA TENGAH	INDONESIA	32	Toko Kelontong	190105	PT. KTRI DISTRIBUSI
5	84029G	234 KERETEK 12	ROKOK	6	3.39	48477	3012739	17850	RISKA CELL ADAM MALIK	JL. ADAM MALIK	SAMARINDA	IANTAN TIMUR	INDONESIA	32	Toko Kelontong	130113	CV. EKA PUTRA
6	84029E	234 KERETEK 16	ROKOK	6	3.39	48477	1921012	17850	ABADI MART	JL.GROWONG - PUCUNG REJO MUNTILAN.	MAGELANG	AWA TENGAH	INDONESIA	42	Mini Market	190105	PT. KTRI DISTRIBUSI
7	22752	26-PROMIL GOLD 400G	TEPUNG	2	7.65	109395	1915415	17850	CV PENI PUTRI	GATEN, CATURTUNGGAL, DEPOK, SLEMAN	SLEMAN	YOGYAKARTA	INDONESIA	32	Toko Kelontong	190104	MBER KASIH PANGAN
8	21730	2TANG AIR MINERAL 240ML	MINUMAN	6	4.25	60775	3012901	17850	TOMMY BERSAUDARA	JL. ADAM MALIK 2	SAMARINDA	IANTAN TIMUR	INDONESIA	32	Toko Kelontong	130113	CV. EKA PUTRA
9	22633	2TANG AIR MINERAL 330ML	MINUMAN	6	1.85	26455	1921073	17850	AMI TOKO	JL. MAGELANG - SECANG PAYAMAN JLN.	MAGELANG	AWA TENGAH	INDONESIA	32	Toko Kelontong	190105	PT. KTRI DISTRIBUSI
10	22632	2TANG AIR MINERAL 600ML	MINUMAN	6	1.85	26455	1510009	17850	ADE	CITALANG	IRWAKARTA	JAWA BARAT	INDONESIA	32	Toko Kelontong	50312	PT. JAYA WIJAYA
11	84879	2TANG SPORT 500ML	MINUMAN	32	1.69	24167	3012941	13047	GONIK	JL. AM. RIFADIN	RTANEGARA	IANTAN TIMUR	INDONESIA	32	Toko Kelontong	130113	CV. EKA PUTRA
12	22745	2TANG TEH BUBUK 50GR VANILLA	TEH & KOPI	6	2.1	30030	1912677	13047	TERUS MAKMUR KENTENG	JL. KENTENG RUKO NO 9	LON PROGO	YOGYAKARTA	INDONESIA	32	Toko Kelontong	190104	MBER KASIH PANGAN
13	22748	2TANG TEH HITAM CELUP 60 BAG	TEH & KOPI	6	2.1	30030	1921249	13047	D MART	YUDAN NO.KA-3 DAN KA-5 SALAKAN JLN.	MAGELANG	AWA TENGAH	INDONESIA	42	Mini Market	190105	PT. KTRI DISTRIBUSI
14	22749	7UP 330ML	SUSU	8	3.75	53625	1510647	13047	TK KOKOM	GG FURNITURE. CIBUNGUR	IRWAKARTA	JAWA BARAT	INDONESIA	32	Toko Kelontong	50312	PT. JAYA WIJAYA
15	22310	888 CALIFORNIA NEW ORANGE	PARFUM	6	1.65	23595	1510692	13047	TK RIDWAN	JL SADANG	IRWAKARTA	JAWA BARAT	INDONESIA	32	Toko Kelontong	50312	PT. JAYA WIJAYA
16	84969	A LICAFE 100G	MINUMAN	6	4.25	60775	1922038	13047	IMAM	NEPAK BULUREJO (DPN SMKN 1)	MAGELANG	IAWA TENGAH	INDONESIA	32	Toko Kelontong	190105	PT. KTRI DISTRIBUSI
17	22623	A LICAFE 300G	MINUMAN	3	4.95	70785	1161623	13047	SUGENG	DESA KARANG MULYA	NAH BUMBU	TAN SELATAN	INDONESIA	32	Toko Kelontong	110129	. BORNEO MERAKSA
18	22622	A&W SARSAPARILA 330ML	MINUMAN	2	9.95	142285	1922807	13047	A&W MART	JL RAYA KANDANGAN JLN.	MANGGUNG	IAWA TENGAH	INDONESIA	32	Toko Kelontong	190105	PT. KTRI DISTRIBUSI
19	21754	A1 PERMEN JAGUNG 150G	PERMEN	3	5.95	85085	4012235	13047	TK. ABDUL	KARANGTANJUNG	NDEGLANG	BANTEN	INDONESIA	32	Toko Kelontong	400307	MITRA ABADI YAKAN
20	21755	A1 SHELLA SENBAI RICE 360G	SNACK	3	5.95	85085	3921705	13047	PARADISE	JL RY BENDO PARE	Kediri	Jawa timur	INDONESIA	32	Toko Kelontong	390112	CV. BACKSPACE
21	21777	AAA ANCOVY FISH SAUCE 750G	MAKANAN KALENG	4	7.95	113685	4011200	13047	SC. BAYNOUNA ARENA FUTSAL	R CILEGON KM 8 KERAMAT WATU SERANG	SERANG	BANTEN	INDONESIA	51	Sport Center	400307	MITRA ABADI YAKAN
22	48187	ABAKUS PERMEN 120G KTK PENSIL	MAKANAN KALENG	4	7.95	113685	4012220	13047	TK. MINAH	KASEMEN	SERANG	BANTEN	INDONESIA	32	Toko Kelontong	400307	MITRA ABADI YAKAN
23	22960	ABAKUS PERMEN 30G KTK BLT	MAKANAN KALENG	6	4.25	60775	1630992	13047	ANDALAN (MOTORIS)	JL. RAYA TUNTANG - MBERAN	SALATIGA	IAWA TENGAH	INDONESIA	32	Toko Kelontong	60315	ELUARGA SEJAHTRA
24	22913	ABAKUS PERMEN BONEKA	MINUMAN	3	4.95	70785	1631972	13047	AMBARWATI (MOTORIS)	AMBARAWA	SEMARANG	IAWA TENGAH	INDONESIA	32	Toko Kelontong	60315	ELUARGA SEJAHTRA
25	22912	ABC ALKALIN 9 VOLT	STATIONERY	3	4.95	70785	4012560	13047	TK. MADURA BAROKAH II	SAMPING PENGADILAN	LEBAK	BANTEN	INDONESIA	32	Toko Kelontong	400307	MITRA ABADI YAKAN
26	22914	ABC AYAM GRENG 135ML	SABUN & SAMPHOO	3	4.95	70785	3922237	13047	BINTANG JAYA	JL DAHLIA	KEDIRI	JAWA TIMUR	INDONESIA	32	Toko Kelontong	390112	CV. BACKSPACE
27	21756	ABC AYAM GRENG 340ML	SABUN & SAMPHOO	3	5.95	85085	3021013	13047	ALTON	SEPINGGAN	Balikpapan	limantan Timur	INDONESIA	32	Toko Kelontong	300106	IGRAH DEZA MANDIRI
28	22728	ABC BLACK GOLD 600ML	BUMBU	24	3.75	53625	2311838	12583	BANK BUKOPIN	JL RAYA CIPTO	CIREBON	JAWA BARAT	INDONESIA	32	Toko Kelontong	230303	PT. SAMYUAN XIEXIE
29	22727	ABC BROWNIES 30S	TEH & KOPI	24	3.75	53625	14210	12583	YUDHI, TK	K PINANGPURA, TUA TUNU, GERUNGGANG	KALPINANG	SKA BELITUNG	INDONESIA	32	Toko Kelontong	370205	CV. IGEDE
30	22726	ABC CUP SELERA PEDAS 65G SEMUR	MINUMAN	12	3.75	53625	10302	12583	LILY, TK	SULAIMAN ARIEF, MASJID JAMIK, RANGKUI	KALPINANG	SKA BELITUNG	INDONESIA	32	Toko Kelontong	370205	CV. IGEDE
31	21724	ABC CUP SELERA PEDAS 65G TOMAT	MINUMAN	12	0.85	12155	3217503	12583	SALMA SLM0401	TANJUNG	BANYUMAS	IAWA TENGAH	INDONESIA	32	Toko Kelontong	320302	V. CITRA BERUSAHA
32	21883	ABC DENT 130G	PASTA & S-GIGI	24	0.65	9295	1111981	12583	LAYAN MAJU BERSAMA,PT(MB.GLUGUR)	K.L.YOS SUDARSO NO.338 - MEDAN	AS SELATAN	ATERA UTARA	INDONESIA	41	Super Market	10209	ADIYATMA LESTARA
33	10002	ABC EXTRA PEDAS 155G	ROKOK	48	0.85	12155	1141022	12583	BMS-TK. GITO	JL. HM RAFFI	NGIN BARAT	KALTENG	INDONESIA	32	Toko Kelontong	110124	T. SEJATI DISTRIBUSI
34	21791	ABC JUICE GUAVA 1LTR	MINUMAN	24	1.25	17875	1112722	12583	ANIS ZAHRATUL	JL. KUIN SELATAN	NJARMASIN	KAL-SEL	INDONESIA	32	Toko Kelontong	110121	RA MAKMUR SELALU
35	21035	ABC JUICE JAMBU 250ML	MINUMAN	18	2.95	42185	136	12583	TK.MAJU 9	PEKKABATA	PAREPARE	VESI SELATAN	INDONESIA	32	Toko Kelontong	90146	ATRA PUTRI FARMASI
36	22326	ABC JUICE JAMBU PREMIUM GOLD 1L	MINYAK GORENG	24	2.95	42185	1151746	12583	KOPERASI PRIMKOPPOL RESORT	JL.CILIK RIWUT SAMPING POLDA	NGKARAYA	NTAN TENGAH	INDONESIA	45	Koperasi	110128	PT. MAMA KOSASIH
37	22629	ABC JUICE MIX PREMIUM GOLD 1L	MINYAK GORENG	24	1.95	27885	1811283	12583	CANDRA DEWATA TK	JL. TIBUNG SARI 59 KWANJI DPS	BADUNG	BALI	INDONESIA	32	Toko Kelontong	80110	DELTA MASYARAKAT
38	22659	ABC JUICE ORANGE 1LTR	MINUMAN	24	1.95	27885	1813078	12583	K-24,APT KEBO IWA	JL. KEBO IWA DPS	DENPASAR	BALI	INDONESIA	14	Apotik I	80110	DELTA MASYARAKAT
39	22631	ABC JUICE ORANGE GOLD 1LTR	MAKANAN	24	1.95	27885	1112246	12583	SOFIA	JL. MALINTANG	BANJAR	KAL-SEL	INDONESIA	75	Salesman Spreading	110121	RA MAKMUR SELALU
40	22661	ABC JUICE SIRSAK	MINUMAN	20	0.85	12155	2211334	12583	CHAMPION FULTSAL ARAYA	PERUM PBI BLOK A-7 BLIMBING	MALANG	JAWA TIMUR	INDONESIA	51	Sport Center	220109	KOTAK AGRO SUBUR
41	21731	ABC JUICE SOY BEAN 1LTR	MINUMAN	24	1.65	23595	1811548	12583	PUTRA MAS,TK	PERUMAHAN SWAMANDALA NO. 9	BADUNG	BALI	INDONESIA	32	Toko Kelontong	80110	DELTA MASYARAKAT
42	22900	ABC JUS APPLE 1000ML	OBATAN	24	2.95	42185	2721737	12583	PRIMKOPPOL AMANAH	JL MH THAMRIN RUKO 9/10	JEMBER		INDONESIA	45	Koperasi	270109	. SENANTIASA ABADI
43	21913	ABC KACANG HIJAU 250ML	MINUMAN	12	3.75	53625	1161439	12583	TOKO EDI	JL MENTEWE SAMPING ASEP	NAH BUMBU	TAN SELATAN	INDONESIA	32	Toko Kelontong	110129	. BORNEO MERAKSA

WHY SUPERVISED LEARNING?

- SUPERVISED LEARNING ALLOWS YOU TO COLLECT DATA OR PRODUCE A DATA OUTPUT FROM THE PREVIOUS EXPERIENCE.
- HELPS YOU TO OPTIMIZE PERFORMANCE CRITERIA USING EXPERIENCE
- SUPERVISED MACHINE LEARNING HELPS YOU TO SOLVE VARIOUS TYPES OF REAL-WORLD COMPUTATION PROBLEMS.

WHY UNSUPERVISED LEARNING?

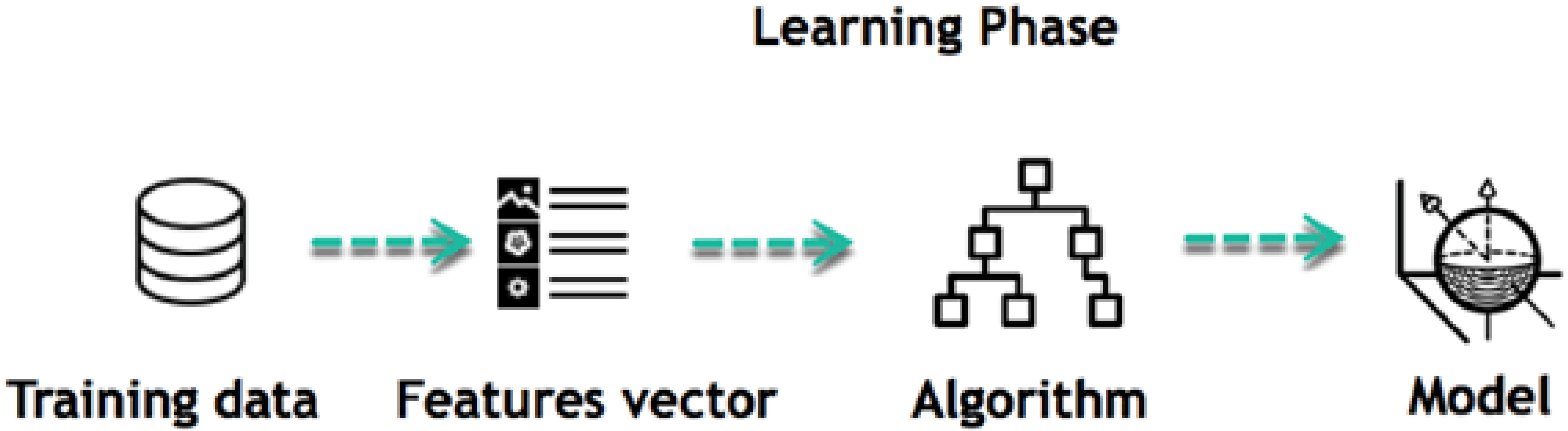
- UNSUPERVISED MACHINE LEARNING FINDS ALL KIND OF UNKNOWN PATTERNS IN DATA.
- UNSUPERVISED METHODS HELP YOU TO FIND FEATURES WHICH CAN BE USEFUL FOR CATEGORIZATION.
- IT IS TAKEN PLACE IN REAL TIME, SO ALL THE INPUT DATA TO BE ANALYZED AND LABELED IN THE PRESENCE OF LEARNERS.
- IT IS EASIER TO GET UNLABELED DATA FROM A COMPUTER THAN LABELED DATA, WHICH NEEDS MANUAL INTERVENTION

HOW SUPERVISED LEARNING WORKS?

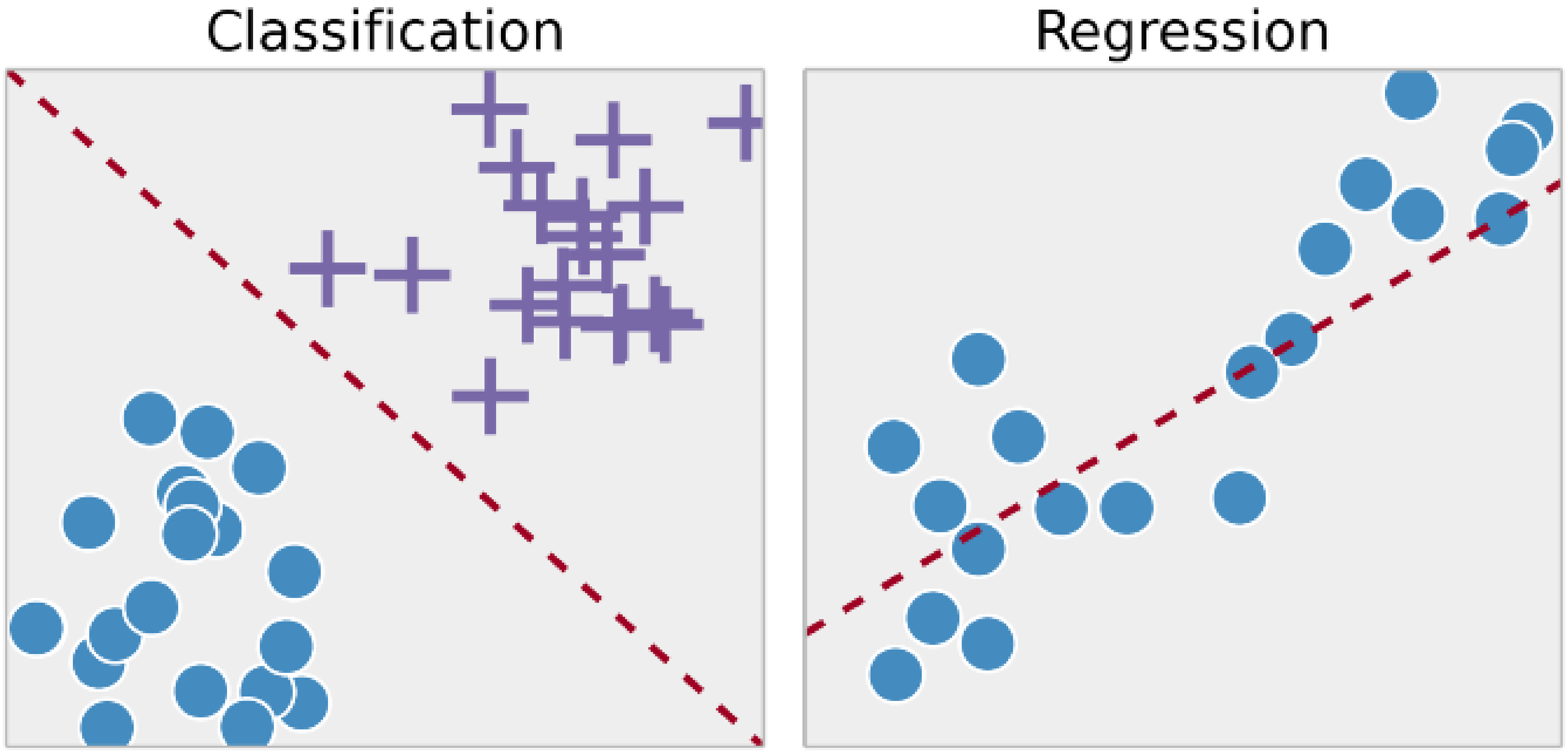
HOW SUPERVISED LEARNING WORKS?

- FOR EXAMPLE, YOU WANT TO TRAIN A MACHINE TO HELP YOU PREDICT HOW LONG IT WILL TAKE YOU TO DRIVE HOME FROM YOUR WORKPLACE. HERE, YOU START BY CREATING A SET OF LABELED DATA. THIS DATA INCLUDES:
  - WEATHER CONDITIONS
  - TIME OF THE DAY
  - HOLIDAYS
- THE FIRST THING YOU REQUIRES TO CREATE IS A TRAINING DATA SET.
- THIS TRAINING SET WILL CONTAIN THE TOTAL COMMUTE TIME AND CORRESPONDING FACTORS LIKE WEATHER, TIME, ETC.
- BASED ON THIS TRAINING SET, YOUR MACHINE MIGHT SEE THERE'S A DIRECT RELATIONSHIP BETWEEN THE AMOUNT OF RAIN AND TIME YOU WILL TAKE TO GET HOME.

LEARNING PHASE



TYPES OF SUPERVISED MACHINE LEARNING TECHNIQUES



CLASSIFICATION:

- CLASSIFICATION MEANS TO GROUP THE OUTPUT INSIDE A CLASS. IF THE ALGORITHM TRIES TO LABEL INPUT INTO TWO DISTINCT CLASSES, IT IS CALLED BINARY CLASSIFICATION.
- SELECTING BETWEEN MORE THAN TWO CLASSES IS REFERRED TO AS MULTICLASS CLASSIFICATION.
- **EXAMPLE**: DETERMINING WHETHER OR NOT SOMEONE WILL BE A DEFAULTER OF THE LOAN.
- **STRENGTHS**: OUTPUTS ALWAYS HAVE A PROBABILISTIC INTERPRETATION, AND THE ALGORITHM CAN BE REGULARIZED TO AVOID OVERFITTING.
- **WEAKNESSES**: LOGISTIC REGRESSION MAY UNDERPERFORM WHEN THERE ARE MULTIPLE OR NON-LINEAR DECISION BOUNDARIES. THIS METHOD IS NOT FLEXIBLE, SO IT DOES NOT CAPTURE MORE COMPLEX RELATIONSHIPS.

REGRESSION:

- REGRESSION TECHNIQUE PREDICTS A SINGLE OUTPUT VALUE USING TRAINING DATA.
- EXAMPLE: YOU CAN USE REGRESSION TO PREDICT THE HOUSE PRICE FROM TRAINING DATA. THE INPUT VARIABLES WILL BE LOCALITY, SIZE OF A HOUSE, ETC.

HOW UNSUPERVISED LEARNING WORKS?

TYPES OF UNSUPERVISED MACHINE LEARNING TECHNIQUES

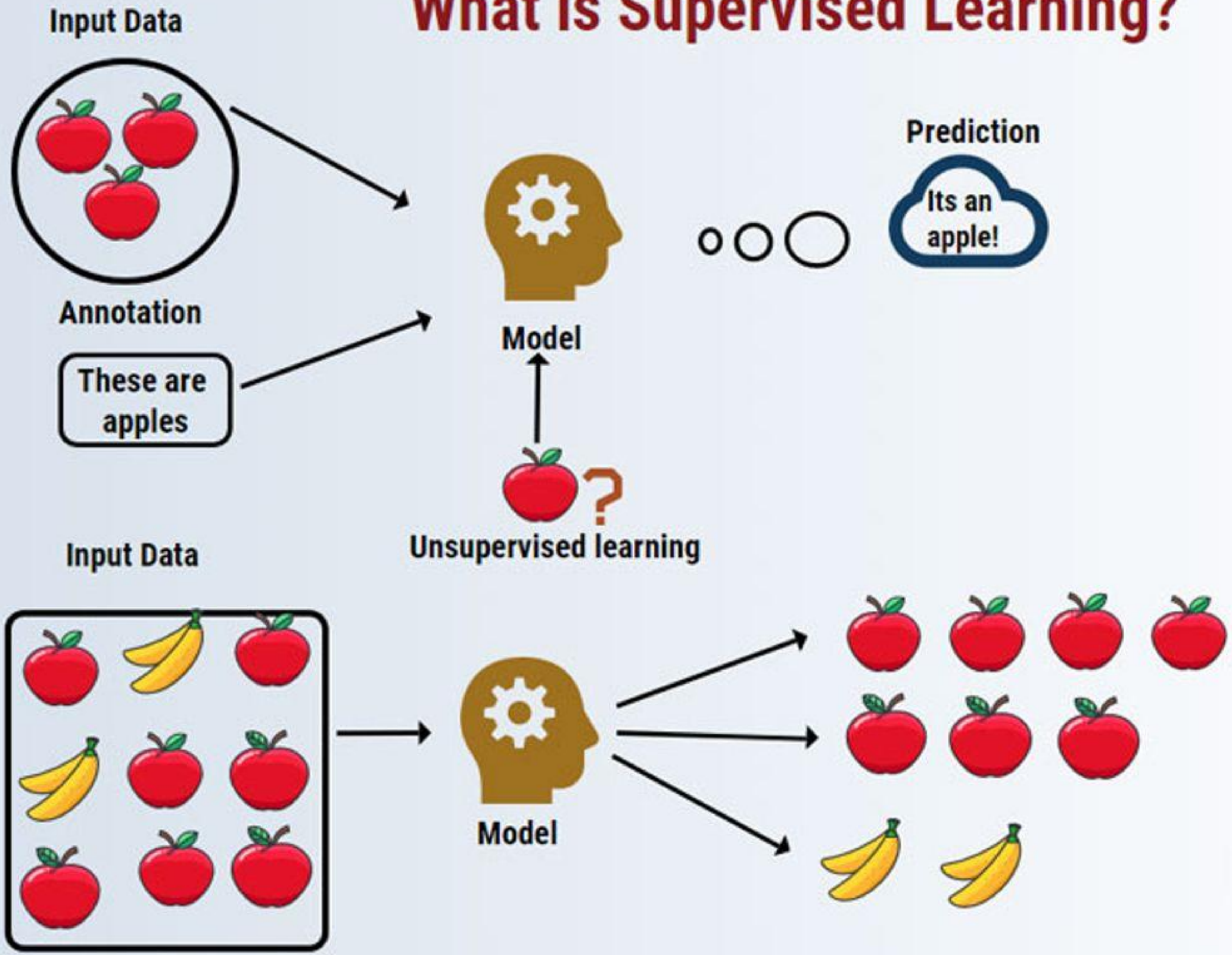
CLUSTERING

- CLUSTERING IS AN IMPORTANT CONCEPT WHEN IT COMES TO UNSUPERVISED LEARNING. IT MAINLY DEALS WITH FINDING A STRUCTURE OR PATTERN IN A COLLECTION OF UNCATEGORIZED DATA.
- CLUSTERING ALGORITHMS WILL PROCESS YOUR DATA AND FIND NATURAL CLUSTERS (GROUPS) IF THEY EXIST IN THE DATA. YOU CAN ALSO MODIFY HOW MANY CLUSTERS YOUR ALGORITHMS SHOULD IDENTIFY. IT ALLOWS YOU TO ADJUST THE GRANULARITY OF THESE GROUPS.

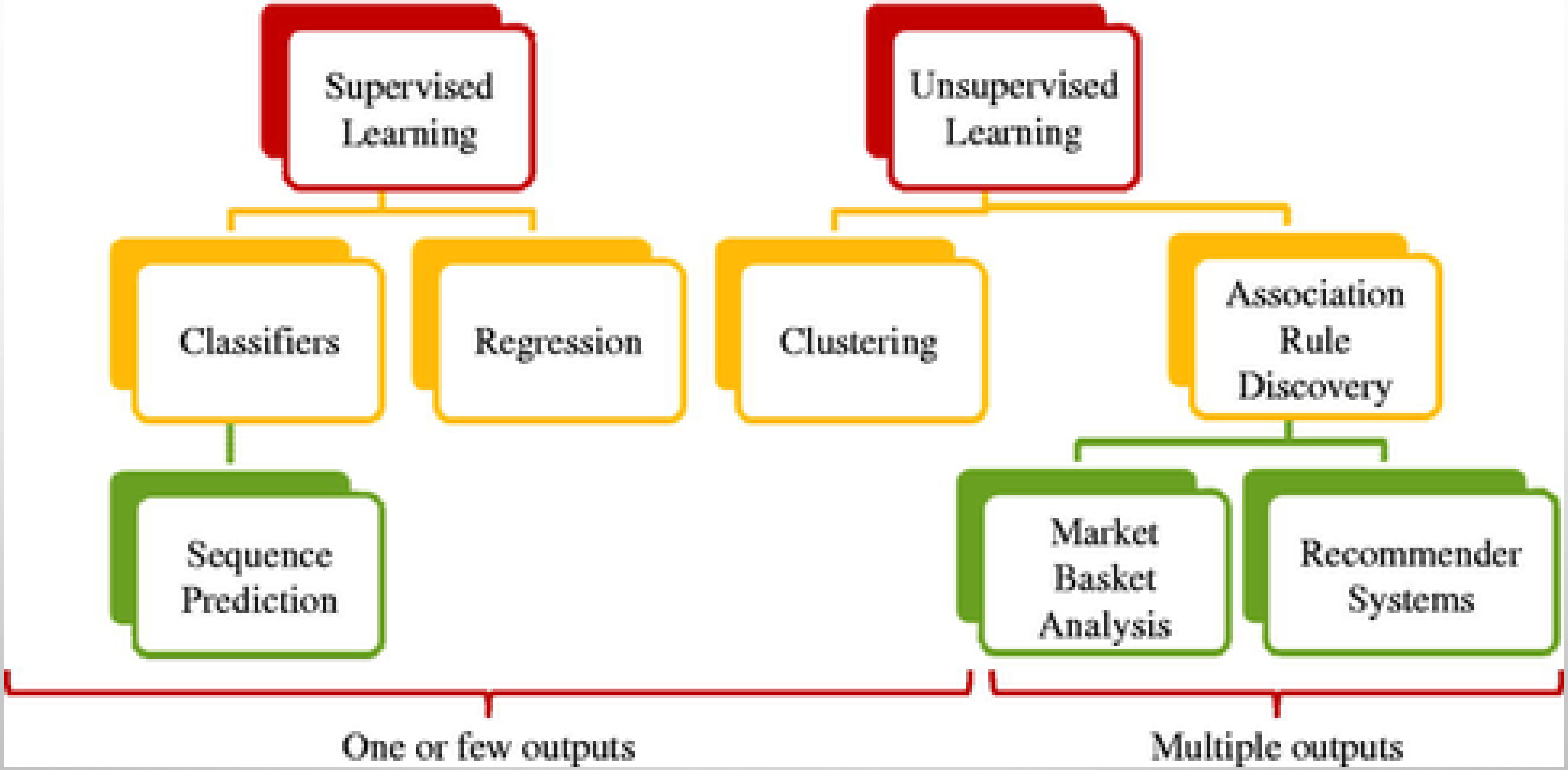
ASSOCIATION

- ASSOCIATION RULES ALLOW YOU TO ESTABLISH ASSOCIATIONS AMONGST DATA OBJECTS INSIDE LARGE DATABASES. THIS UNSUPERVISED TECHNIQUE IS ABOUT DISCOVERING EXCITING RELATIONSHIPS BETWEEN VARIABLES IN LARGE DATABASES.
- FOR EXAMPLE, PEOPLE THAT BUY A NEW HOME MOST LIKELY TO BUY NEW FURNITURE.
- A SUBGROUP OF CANCER PATIENTS GROUPED BY THEIR GENE EXPRESSION MEASUREMENTS
- GROUPS OF SHOPPER BASED ON THEIR BROWSING AND PURCHASING HISTORIES
- MOVIE GROUP BY THE RATING GIVEN BY MOVIES VIEWERS

What is Supervised Learning?

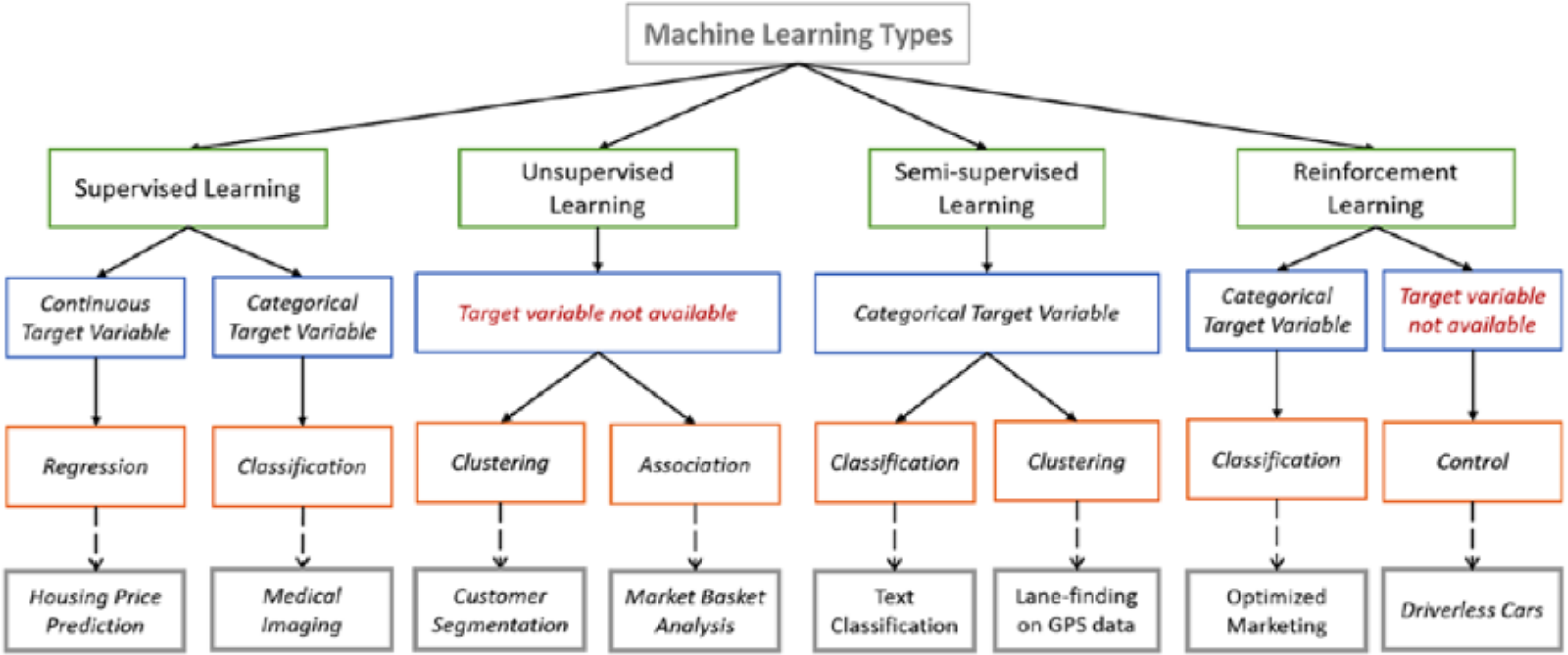


SUPERVISED VS. UNSUPERVISED LEARNING


Supervised Learning
Unsupervised Learning
Classifiers
Regression
Clustering
Association Rule Discovery
Sequence Prediction
Market Basket Analysis
Recommender Systems
One or few outputs
Multiple outputs

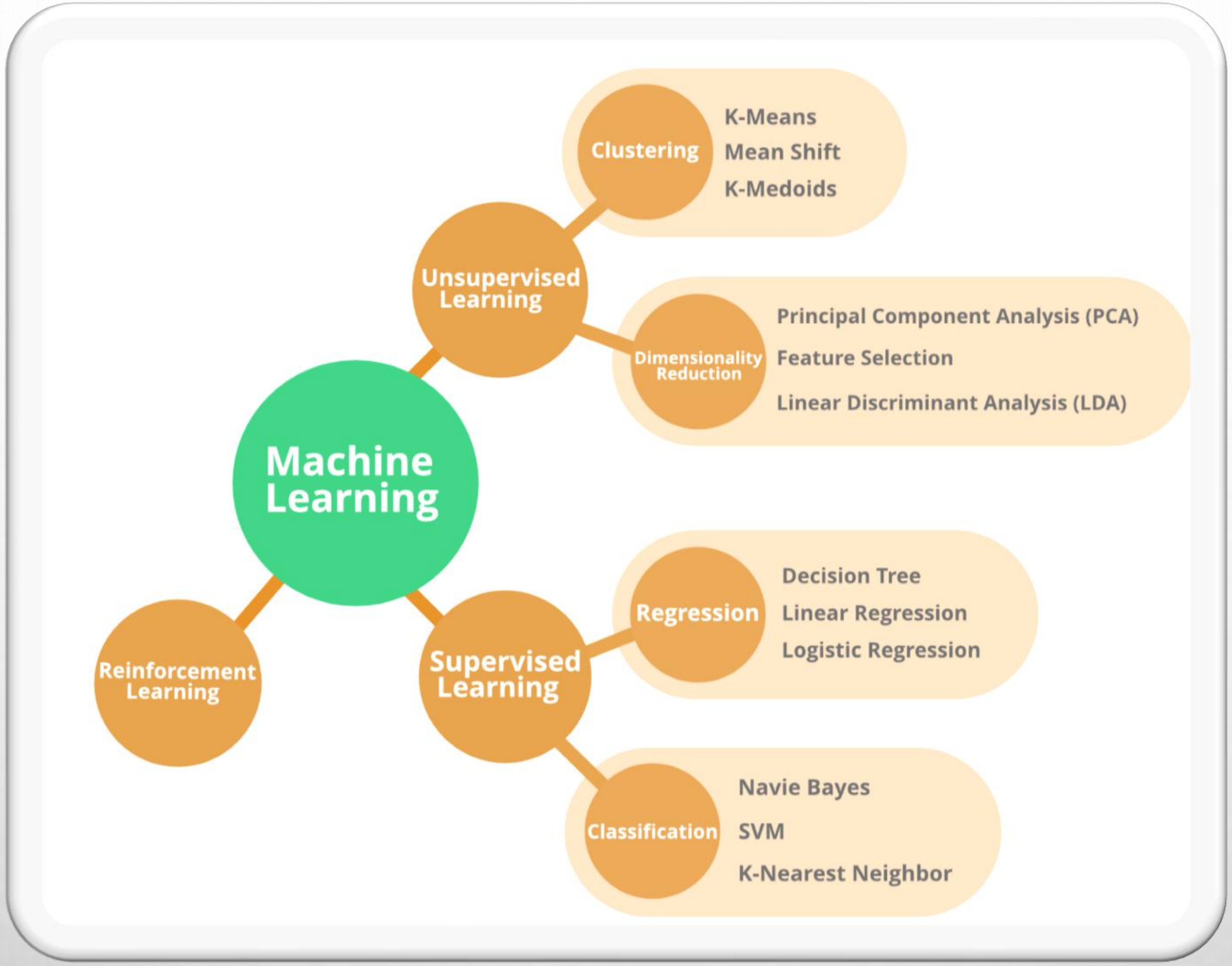
Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Process	In a supervised learning model, input and output variables will be given.	In unsupervised learning model, only input data will be given
Input Data	Algorithms are trained using labeled data.	Algorithms are used against data which is not labeled
Algorithms Used	Support vector machine, Neural network, Linear and logistics regression, random forest, and Classification trees.	Unsupervised algorithms can be divided into different categories: like Cluster algorithms, K-means, Hierarchical clustering, etc.
Computational Complexity	Supervised learning is a simpler method.	Unsupervised learning is computationally complex
Use of Data	Supervised learning model uses training data to learn a link between the input and the outputs.	Unsupervised learning does not use output data.
Accuracy of Results	Highly accurate and trustworthy method.	Less accurate and trustworthy method.
Real Time Learning	Learning method takes place offline.	Learning method takes place in real time.
Number of Classes	Number of classes is known.	Number of classes is not known.
Main Drawback	Classifying big data can be a real challenge in Supervised Learning.	You cannot get precise information regarding data sorting, and the output as data used in unsupervised learning is labeled and not known.

	Supervised	Unsupervised
Pattern identification	Data Scientist identifies patterns in the dataset and creates a model for prediction	Let the machine discover new patterns from the dataset
Pre-requisite	Some data is needed (obviously, more is better)	Require very large datasets
When to use	Used when labeled data is available (i.e., historical dataset with label to be predicted is available e.g. Tip amount and Bill amount for all rides in the past 12 months)	Used when labeled data is not available
Sample use case	• Credit card fraud detection based on past fraudulent transactions (flag indicating fraud in the historical data is the labeled data) – uses a Classification algorithm • Predict Tip amount based on Bill amount – uses Linear regression	• Discover groups of similar visitors to a web site (we don't tell the algorithm which group a visitor belongs to, it finds those connections without our help) – uses Clustering algorithm • Discover product affinity to identify which two products best sells together – uses Association Rule Learning
Algorithm types	• Regression (for continuous label - e.g. tip amount) • Classification aka Logistic regression (for discrete or categorical value – e.g. Female/Male, Yes/No/May be) • Support Vector Machines (SVM) • k-Nearest neighbors • Decision trees and Random forests	• Clustering o k-Means, Hierarchical Cluster Analysis • Visualization and dimensionality reduction • Association rule learning

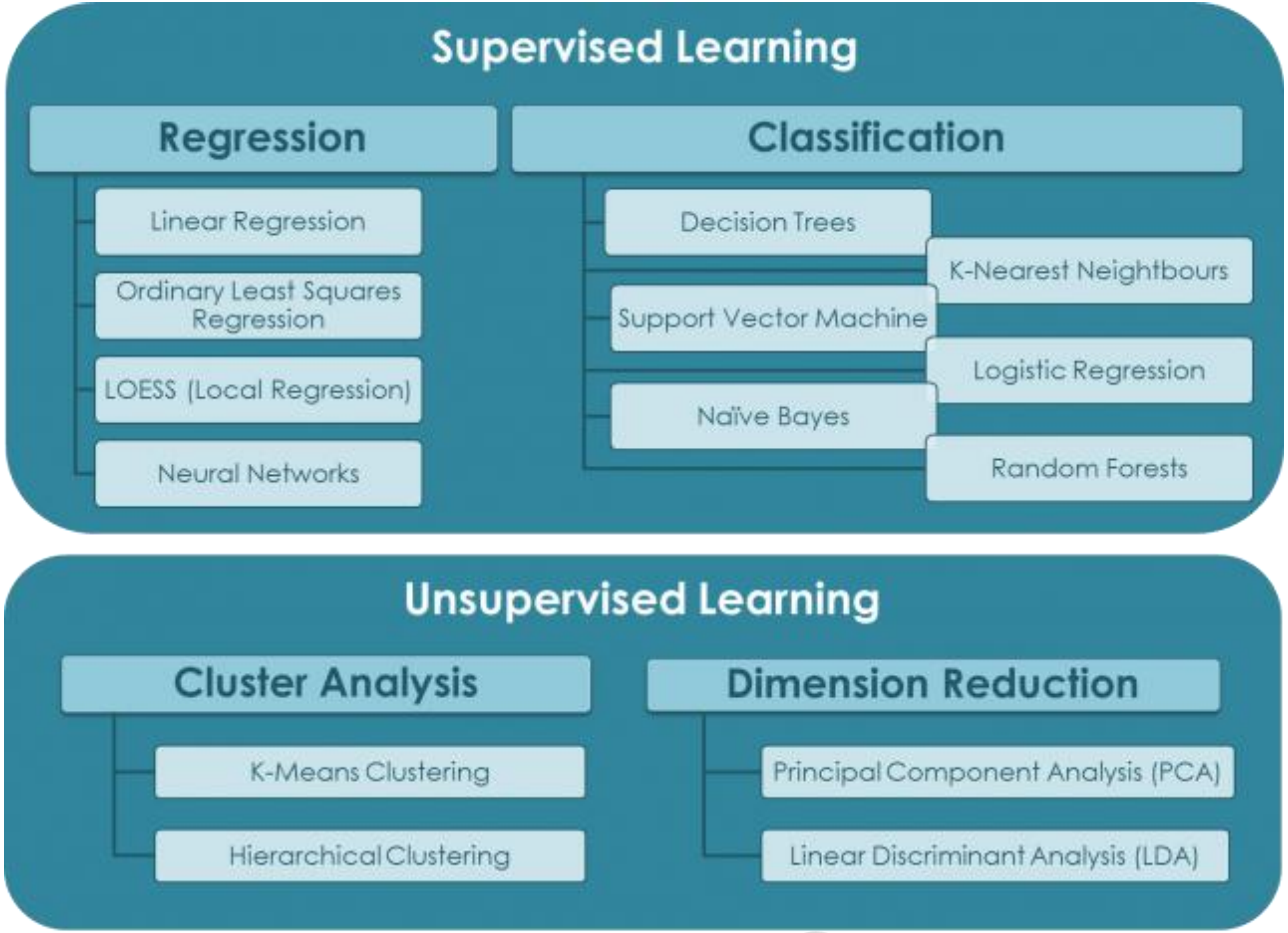

Machine Learning Types
Supervised Learning
Unsupervised Learning
Semi-supervised Learning
Reinforcement Learning
Continuous Target Variable
Categorical Target Variable
Target variable not available
Categorical Target Variable
Categorical Target Variable
Target variable not available
Regression
Classification
Clustering
Association
Classification
Clustering
Classification
Control
Housing Price Prediction
Medical Imaging
Customer Segmentation
Market Basket Analysis
Text Classification
Lane-finding on GPS data
Optimized Marketing
Driverless Cars

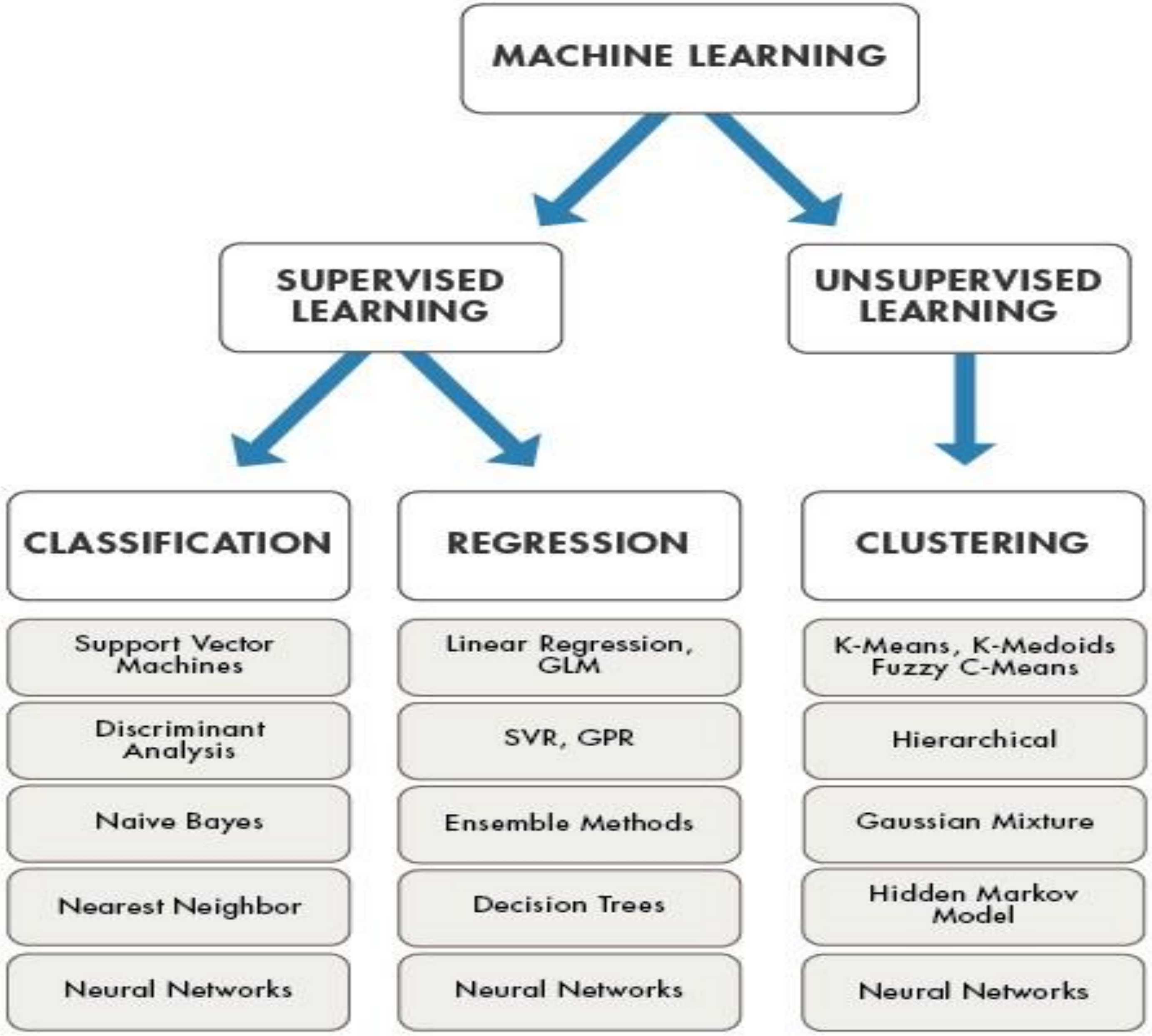
Styles of Learning

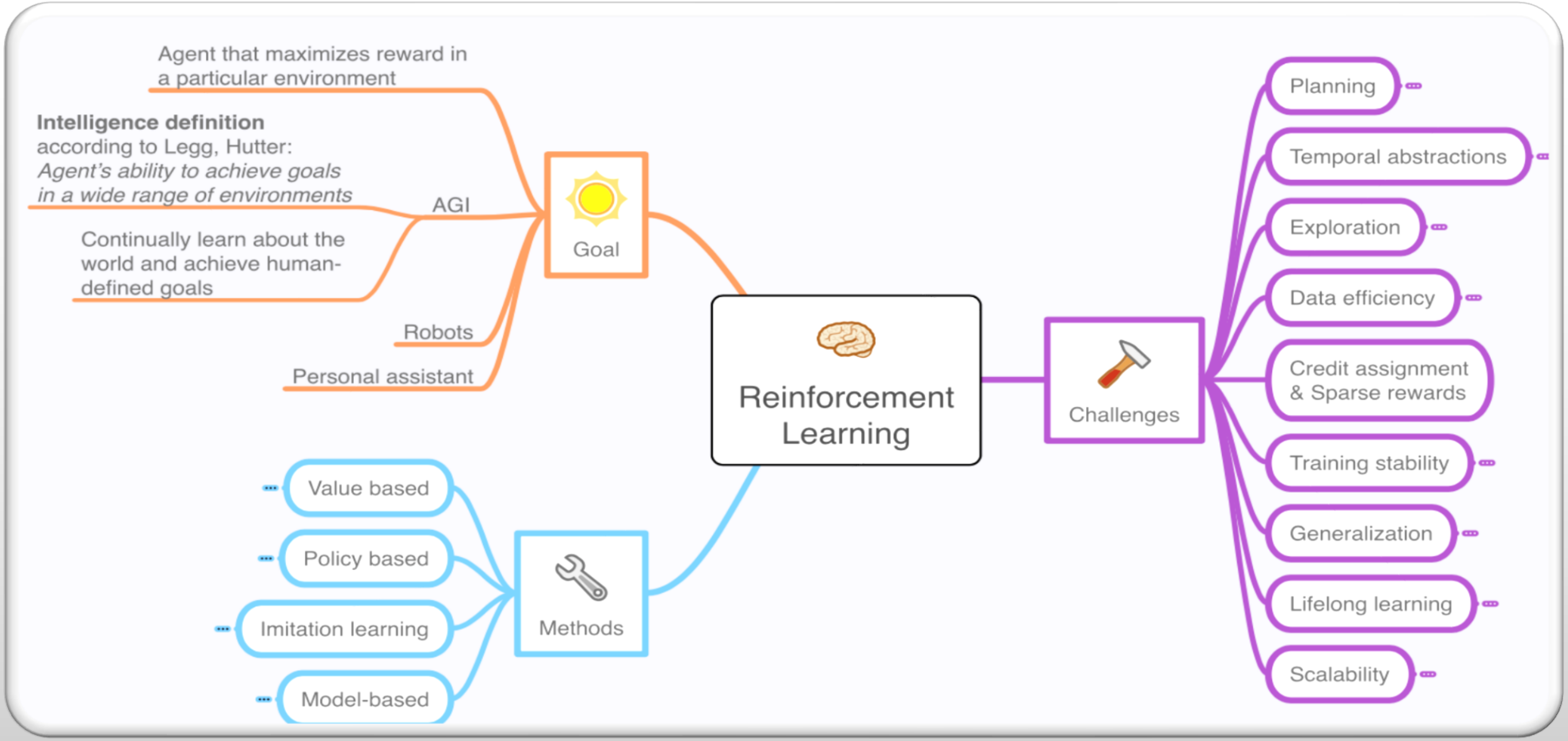
Supervised	Unsupervised	Semi-Supervised	Reinforcement
• Data has **known labels** or output	• Labels or output unknown • Focus on **finding patterns and gaining insight** from the data	• Labels or output known for **a subset of** data • A blend of supervised and unsupervised learning	• Focus on **making decisions** based on previous experience • Policy-making with feedback
• Insurance underwriting • Fraud detection	• Customer clustering • Association rule mining	• Medical predictions (where tests and expert diagnoses are expensive, and only part of the population receives them)	• Game AI • Complex decision problems • Reward systems



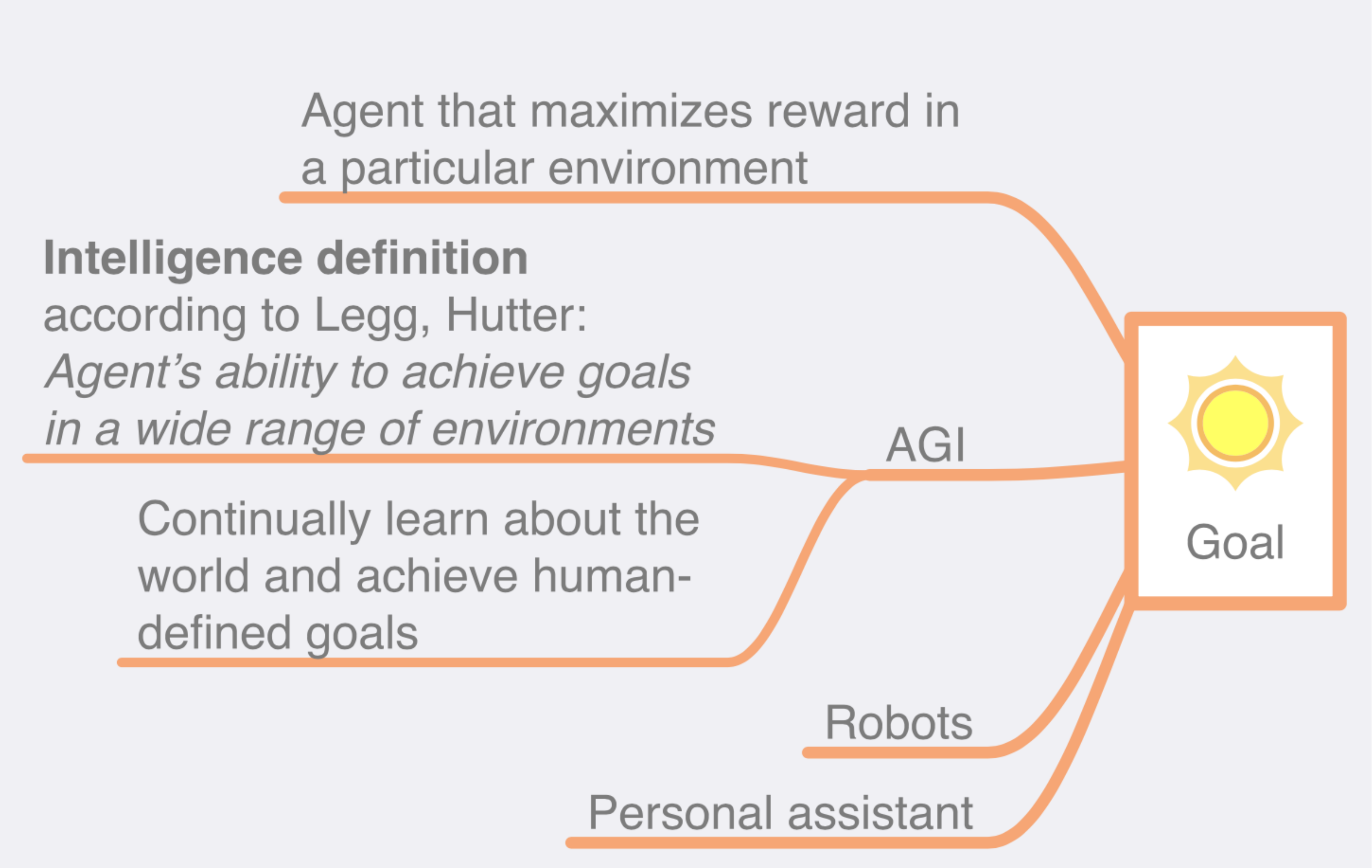
Machine Learning
Unsupervised Learning
Clustering
K-Means
Mean Shift
K-Medoids
Dimensionality Reduction
Principal Component Analysis (PCA)
Feature Selection
Linear Discriminant Analysis (LDA)
Supervised Learning
Regression
Decision Tree
Linear Regression
Logistic Regression
Classification
Navie Bayes
SVM
K-Nearest Neighbor
Reinforcement Learning

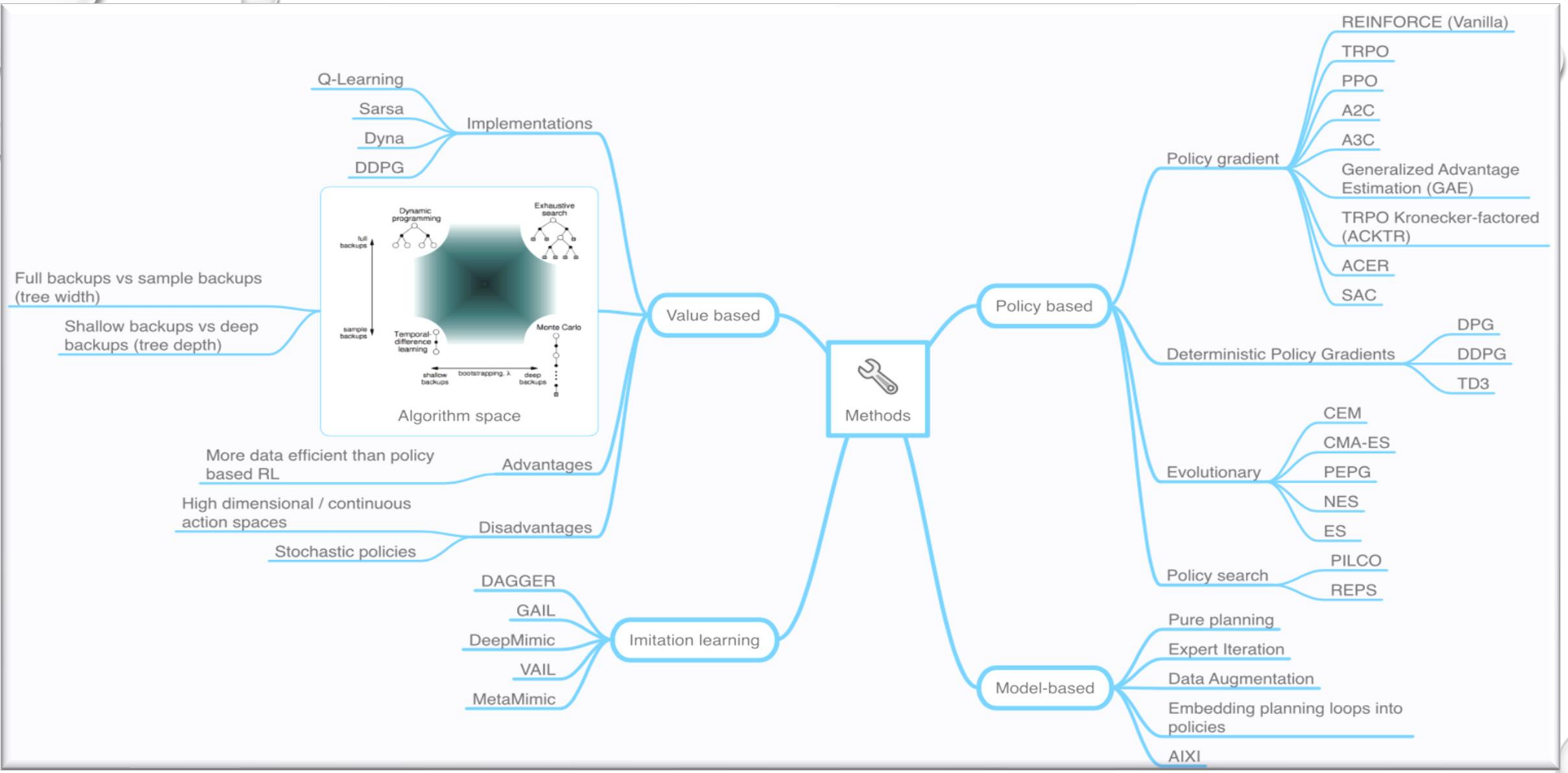

Supervised Learning
Regression
Linear Regression
Ordinary Least Squares Regression
LOESS (Local Regression)
Neural Networks
Classification
Decision Trees
K-Nearest Neighbours
Support Vector Machine
Logistic Regression
Naïve Bayes
Random Forests
Unsupervised Learning
Cluster Analysis
K-Means Clustering
Hierarchical Clustering
Dimension Reduction
Principal Component Analysis (PCA)
Linear Discriminant Analysis (LDA)

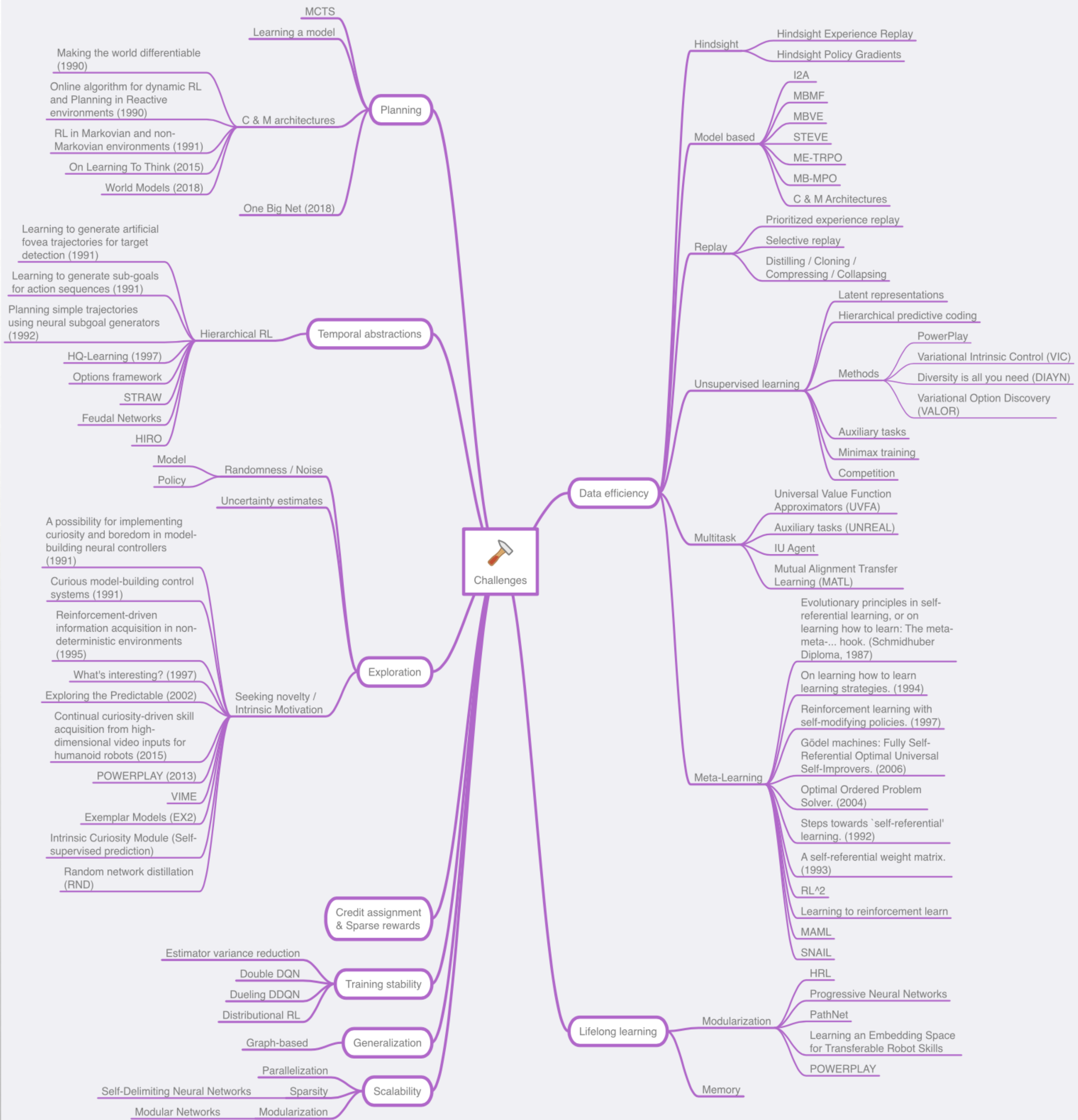

MACHINE LEARNING
SUPERVISED LEARNING
UNSUPERVISED LEARNING
CLASSIFICATION
REGRESSION
CLUSTERING
Support Vector Machines
Discriminant Analysis
Naive Bayes
Nearest Neighbor
Neural Networks
Linear Regression, GLM
SVR, GPR
Ensemble Methods
Decision Trees
Neural Networks
K-Means, K-Medoids Fuzzy C-Means
Hierarchical
Gaussian Mixture
Hidden Markov Model
Neural Networks



REINFORCEMENT LEARNING


Agent that maximizes reward in a particular environment
Intelligence definition according to Legg, Hutter: *Agent's ability to achieve goals in a wide range of environments*
AGI
Continually learn about the world and achieve human-defined goals
Robots
Personal assistant
Goal


Methods
Value based
Implementations
Q-Learning
Sarsa
Dyna
DDPG
Dynamic programming
Exhaustive search
full backups
sample backups
Temporal-difference learning
Monte Carlo
shallow backups
bootstrapping, λ
deep backups
Algorithm space
Full backups vs sample backups (tree width)
Shallow backups vs deep backups (tree depth)
Advantages
More data efficient than policy based RL
Disadvantages
High dimensional / continuous action spaces
Stochastic policies
Imitation learning
DAGGER
GAIL
DeepMimic
VAIL
MetaMimic
Policy based
Policy gradient
REINFORCE (Vanilla)
TRPO
PPO
A2C
A3C
Generalized Advantage Estimation (GAE)
TRPO Kronecker-factored (ACKTR)
ACER
SAC
Deterministic Policy Gradients
DPG
DDPG
TD3
Evolutionary
CEM
CMA-ES
PEPG
NES
ES
Policy search
PILCO
REPS
Model-based
Pure planning
Expert Iteration
Data Augmentation
Embedding planning loops into policies
AIXI


Challenges
Planning
MCTS
Learning a model
C & M architectures
Making the world differentiable (1990)
Online algorithm for dynamic RL and Planning in Reactive environments (1990)
RL in Markovian and non-Markovian environments (1991)
On Learning To Think (2015)
World Models (2018)
One Big Net (2018)
Temporal abstractions
Hierarchical RL
Learning to generate artificial fovea trajectories for target detection (1991)
Learning to generate sub-goals for action sequences (1991)
Planning simple trajectories using neural subgoal generators (1992)
HQ-Learning (1997)
Options framework
STRAW
Feudal Networks
HIRO
Exploration
Randomness / Noise
Model
Policy
Uncertainty estimates
Seeking novelty / Intrinsic Motivation
A possibility for implementing curiosity and boredom in model-building neural controllers (1991)
Curious model-building control systems (1991)
Reinforcement-driven information acquisition in non-deterministic environments (1995)
What's interesting? (1997)
Exploring the Predictable (2002)
Continual curiosity-driven skill acquisition from high-dimensional video inputs for humanoid robots (2015)
POWERPLAY (2013)
VIME
Exemplar Models (EX2)
Intrinsic Curiosity Module (Self-supervised prediction)
Random network distillation (RND)
Credit assignment & Sparse rewards
Training stability
Estimator variance reduction
Double DQN
Dueling DDQN
Distributional RL
Generalization
Graph-based
Scalability
Parallelization
Sparsity
Self-Delimiting Neural Networks
Modularization
Modular Networks
Data efficiency
Hindsight
Hindsight Experience Replay
Hindsight Policy Gradients
Model based
I2A
MBMF
MBVE
STEVE
ME-TRPO
MB-MPO
C & M Architectures
Replay
Prioritized experience replay
Selective replay
Distilling / Cloning / Compressing / Collapsing
Unsupervised learning
Latent representations
Hierarchical predictive coding
Methods
PowerPlay
Variational Intrinsic Control (VIC)
Diversity is all you need (DIAYN)
Variational Option Discovery (VALOR)
Auxiliary tasks
Minimax training
Competition
Multitask
Universal Value Function Approximators (UVFA)
Auxiliary tasks (UNREAL)
IU Agent
Mutual Alignment Transfer Learning (MATL)
Meta-Learning
Evolutionary principles in self-referential learning, or on learning how to learn: The meta-meta-... hook. (Schmidhuber Diploma, 1987)
On learning how to learn learning strategies. (1994)
Reinforcement learning with self-modifying policies. (1997)
Gödel machines: Fully Self-Referential Optimal Universal Self-Improvers. (2006)
Optimal Ordered Problem Solver. (2004)
Steps towards `self-referential' learning. (1992)
A self-referential weight matrix. (1993)
RL^2
Learning to reinforcement learn
MAML
SNAIL
Lifelong learning
Modularization
HRL
Progressive Neural Networks
PathNet
Learning an Embedding Space for Transferable Robot Skills
POWERPLAY
Memory

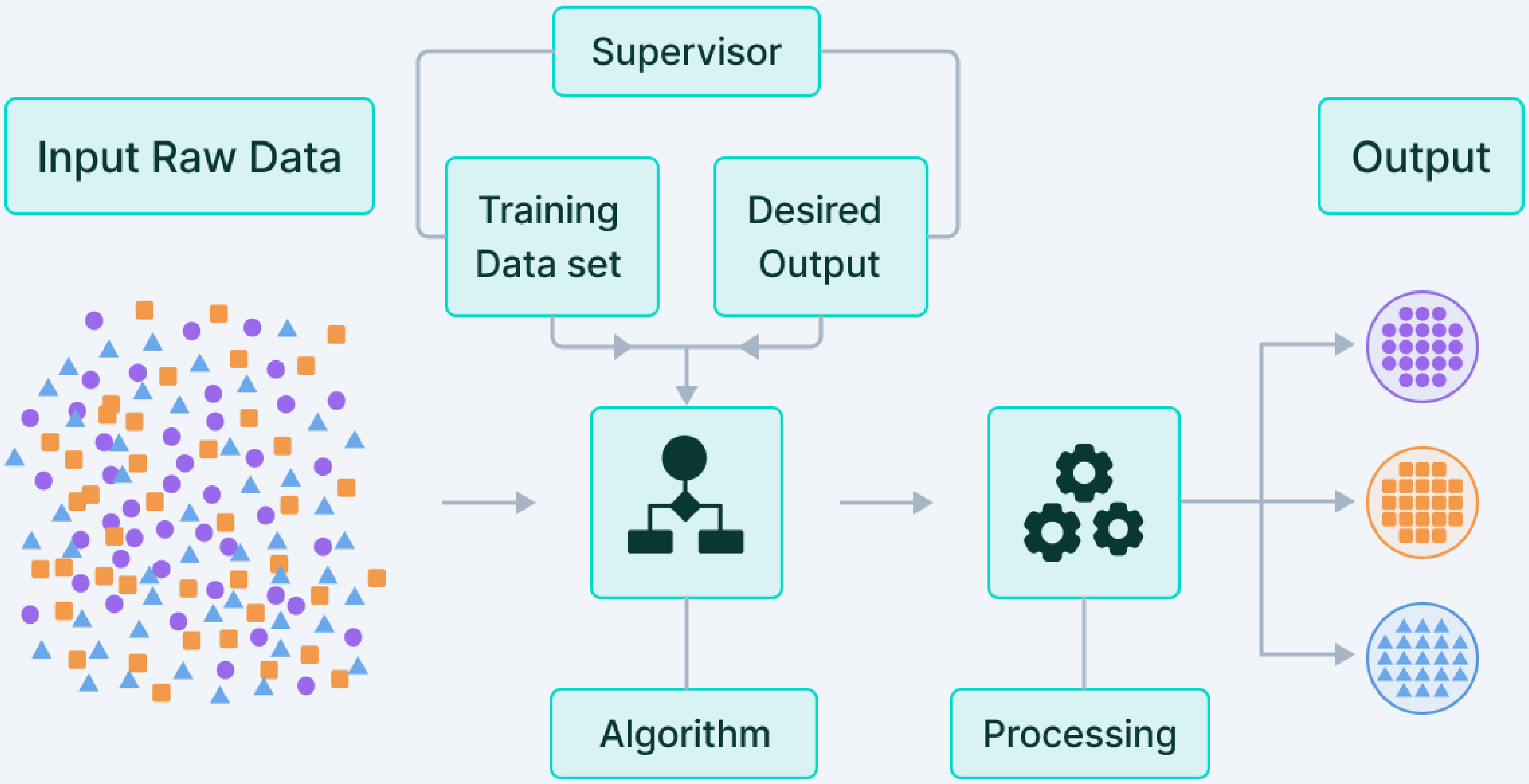
SUMMARY

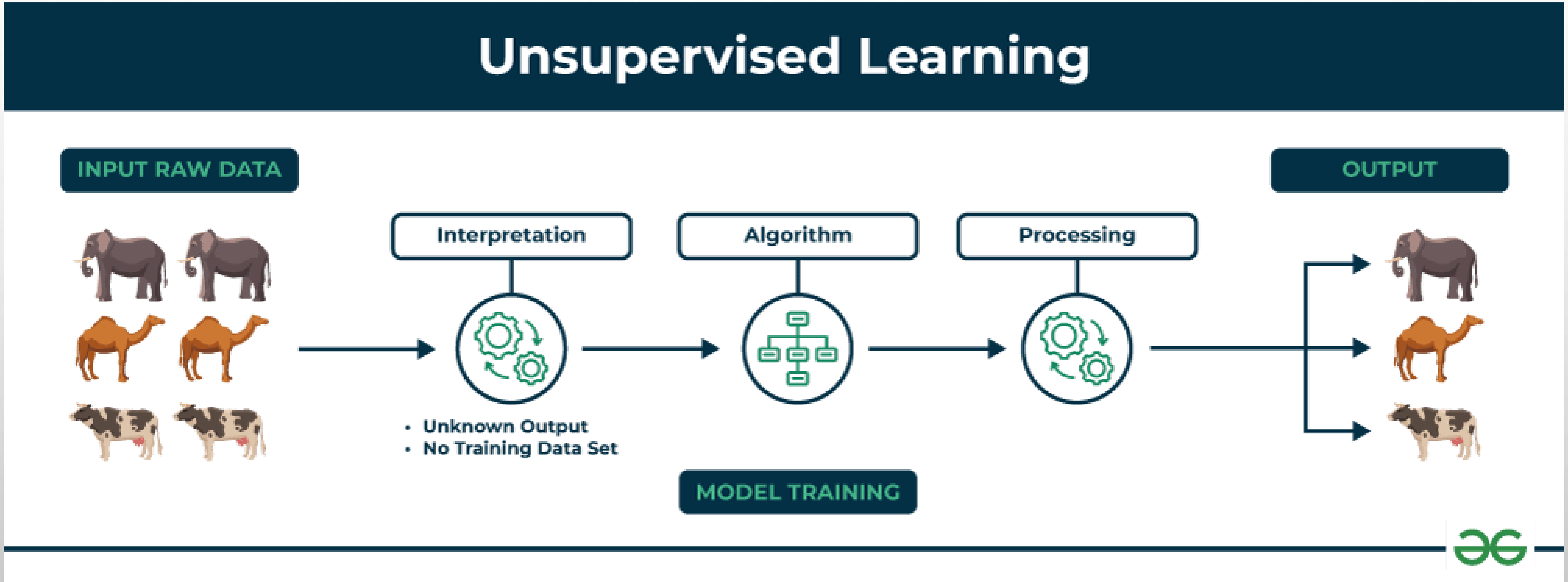
- IN SUPERVISED LEARNING, YOU TRAIN THE MACHINE USING DATA WHICH IS WELL "LABELED."
- UNSUPERVISED LEARNING IS A MACHINE LEARNING TECHNIQUE, WHERE YOU DO NOT NEED TO SUPERVISE THE MODEL.
- SUPERVISED LEARNING ALLOWS YOU TO COLLECT DATA OR PRODUCE A DATA OUTPUT FROM THE PREVIOUS EXPERIENCE.
- UNSUPERVISED MACHINE LEARNING HELPS YOU TO FINDS ALL KIND OF UNKNOWN PATTERNS IN DATA.

SUMMARY

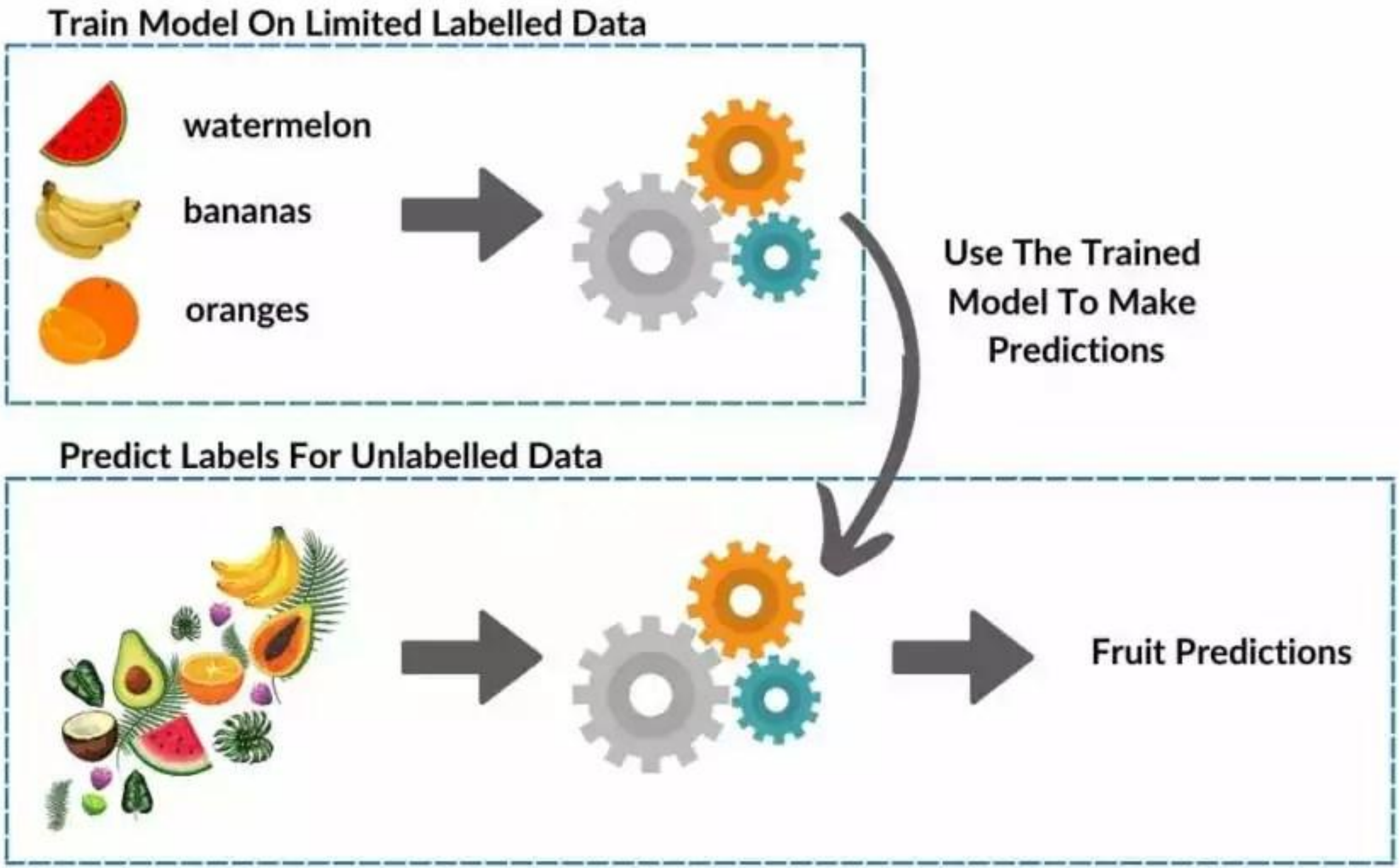
- FOR EXAMPLE, YOU WILL ABLE TO DETERMINE THE TIME TAKEN TO REACH BACK COME BASE ON WEATHER CONDITION, TIMES OF THE DAY AND HOLIDAY.
- FOR EXAMPLE, BABY CAN IDENTIFY OTHER DOGS BASED ON PAST SUPERVISED LEARNING.
- REGRESSION AND CLASSIFICATION ARE TWO TYPES OF SUPERVISED MACHINE LEARNING TECHNIQUES.
- CLUSTERING AND ASSOCIATION ARE TWO TYPES OF UNSUPERVISED LEARNING.
- IN A SUPERVISED LEARNING MODEL, INPUT AND OUTPUT VARIABLES WILL BE GIVEN WHILE WITH UNSUPERVISED LEARNING MODEL, ONLY INPUT DATA WILL BE GIVEN

Supervised Learning




Unsupervised Learning
INPUT RAW DATA
OUTPUT
Interpretation
Algorithm
Processing
• Unknown Output
• No Training Data Set
MODEL TRAINING

How To Expand A Partially-Labelled Dataset With Semi-Supervised Learning



Reinforcement Learning in ML


Thank you